

Advanced Prompting Strategies

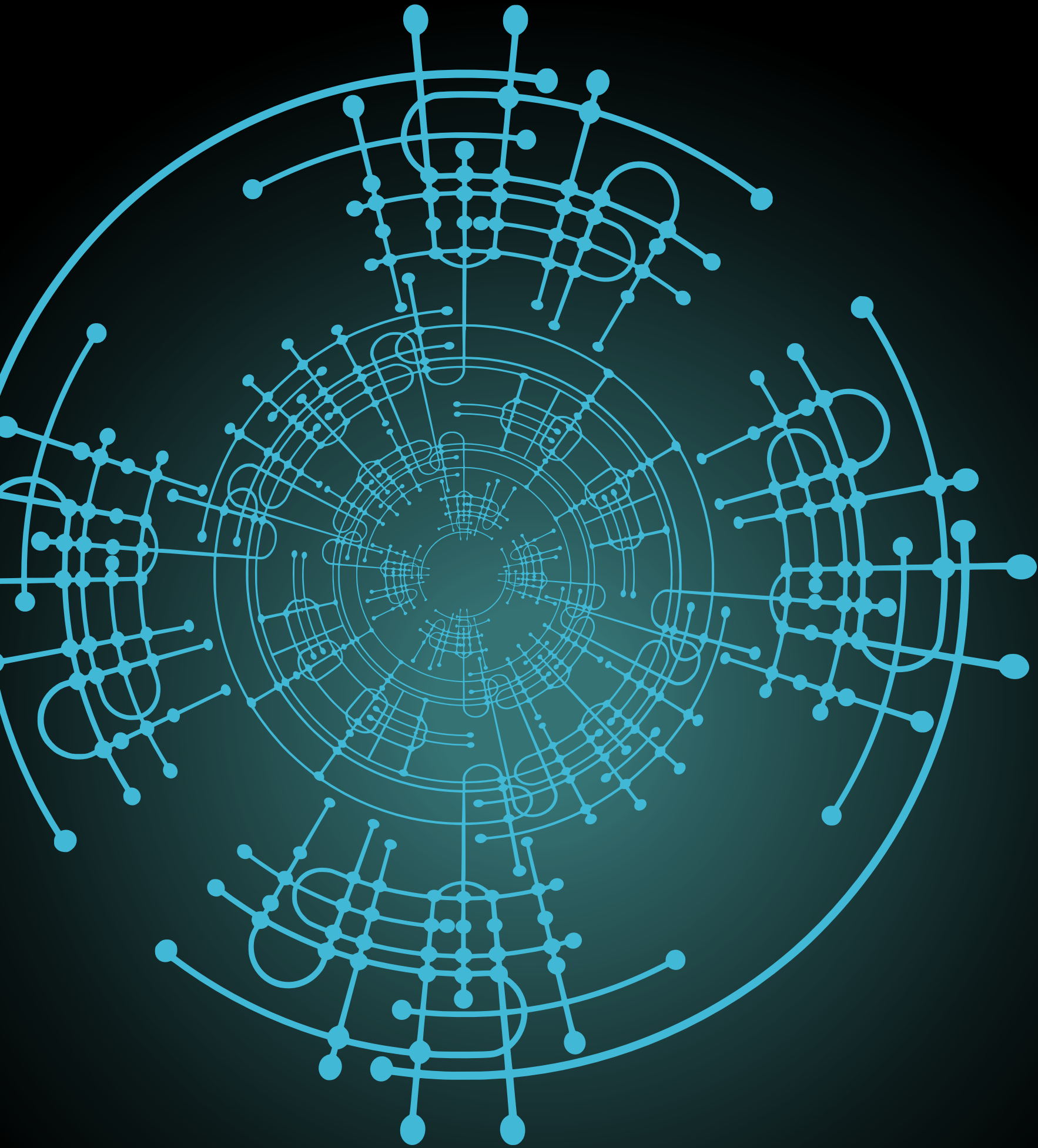
Exploring **Chain of Thought (CoT)** and
Tree of Thoughts (ToT) Reasoning

Presented by: **Aiman Khan**

Why Advanced Prompting?

- Basic prompts give surface-level answers.
- Complex problems need structured reasoning.
- CoT and ToT help AI “think before answering.”
- Used in math, planning, decision-making, and creative problem solving.





- A prompting method that makes AI reason step-by-step.
- Instead of jumping to the final answer, it explains its logic first.
- Helps the model break down complex tasks into small steps.

What is Chain of Thought (CoT)?

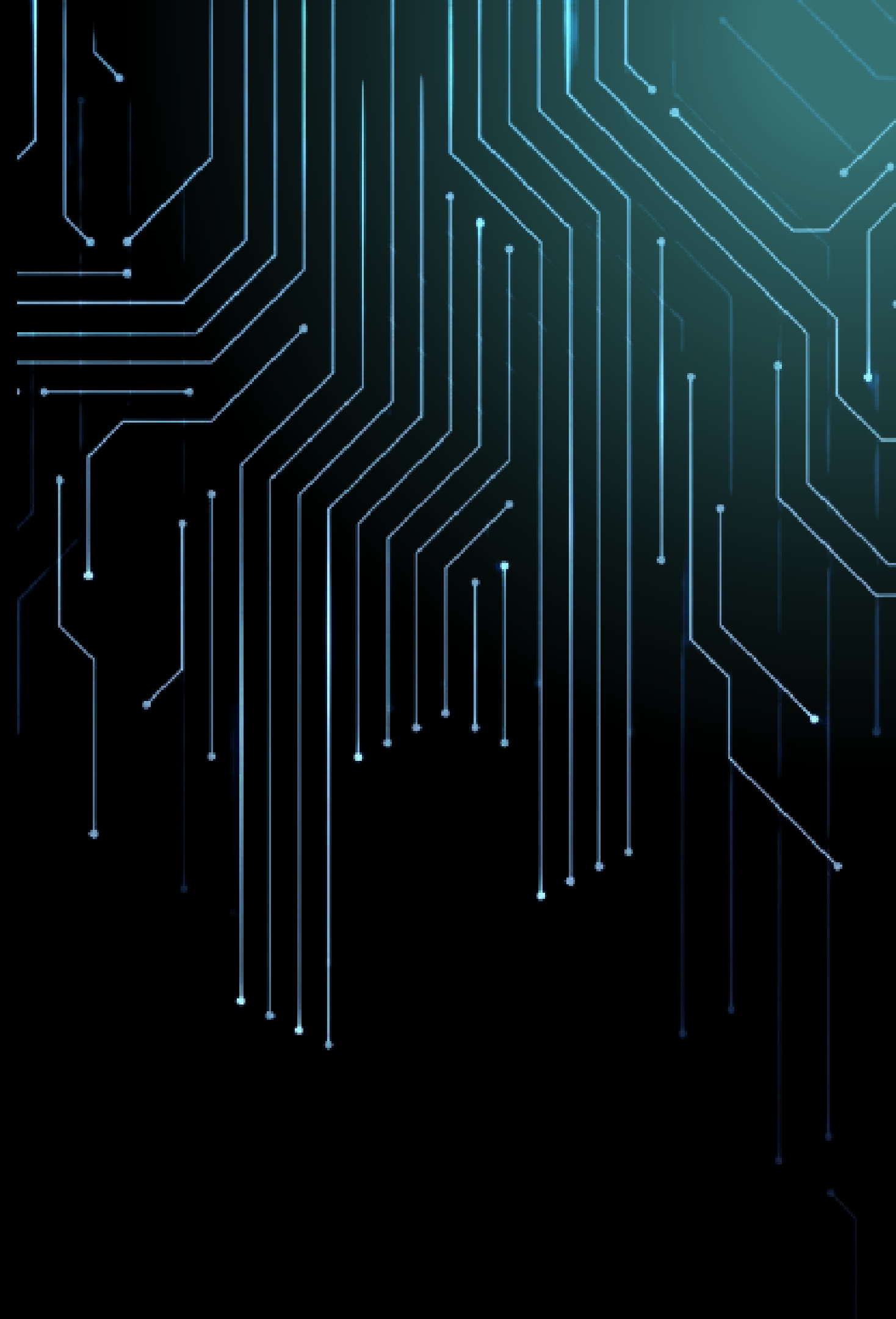
Prompt:

"If I was 6 when my sister was half my age, how old is my sister when I'm 40? Let's solve this step-by-step."

AI Reasoning (Step-by-Step):

1. When I was 6, my sister was 3.
2. Difference in age = 3 years.
3. When I'm 40, my sister = $40 - 3 = 37$.
4. ☒ Answer: 37 years old.

Example of CoT



When to Use CoT



Best for:

- Math problems
- Logical reasoning
- Complex analysis
- Multi-step decision processes



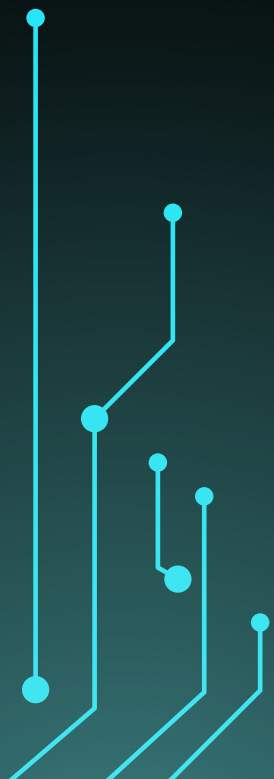
Best Practices:

- Always use phrases like "Let's think step by step."
- Keep temperature = 0 for stable logic.
- Ask model to separate reasoning from the final answer.

- A reasoning framework where AI explores multiple possible paths (branches).
- Each branch = a different approach or idea.
- The AI then evaluates, scores, and synthesizes the best result.



What is Tree of Thoughts (ToT)?



ToT Concept Visualization

Question →

↳ Branch 1 → Explore idea → Score

↳ Branch 2 → Explore idea →

Score

↳ Branch 3 → Explore idea →

Score

→ Synthesize best result 🌟

Each branch represents a different way of solving the problem.



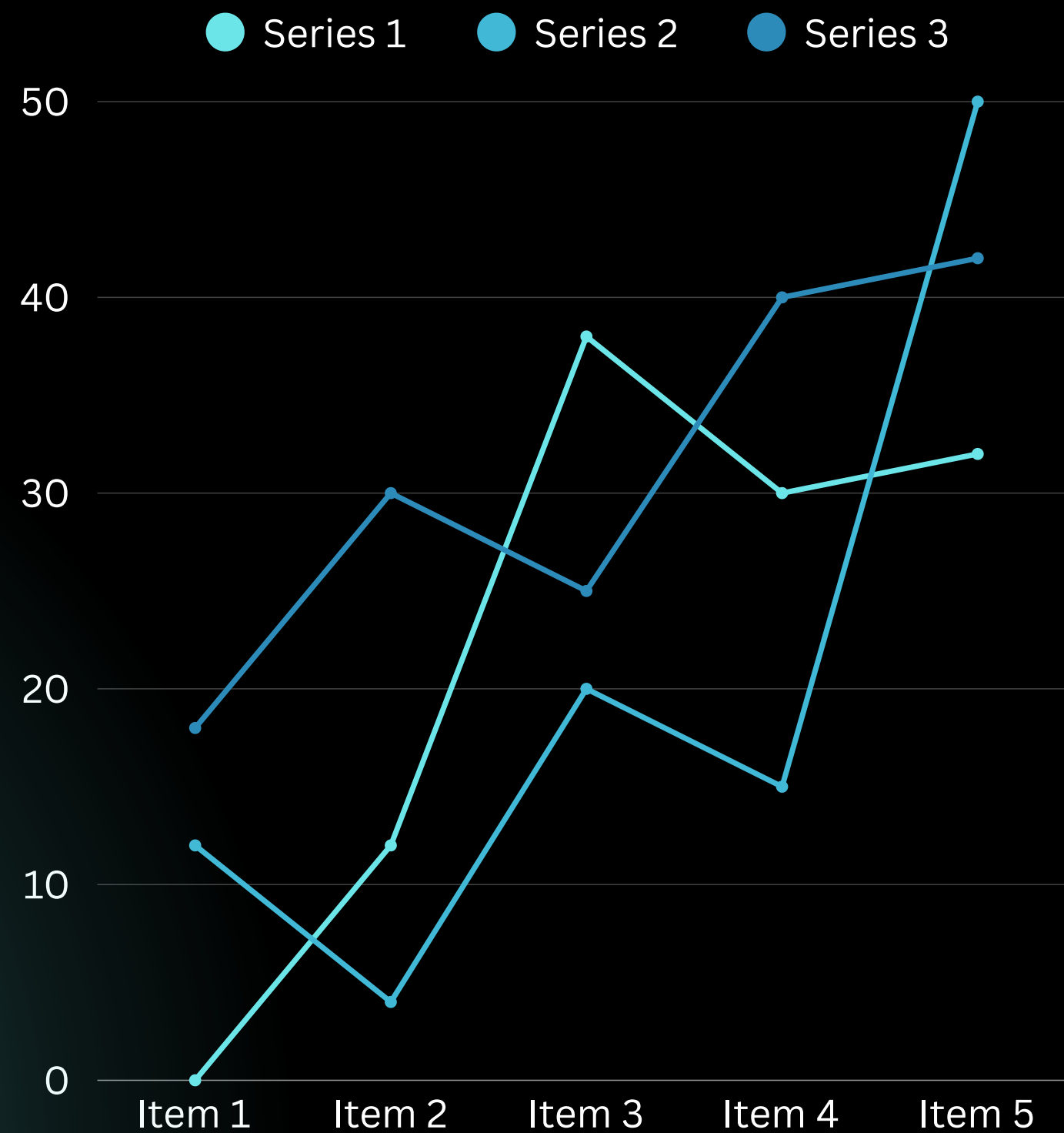
ToT in Action

Scenario: Launching an eco-friendly clothing brand.

AI explores 3 reasoning branches:

1. 🎯 Social Media Campaigns
2. 🌍 Sustainable Pop-up Events
3. 🤝 Collaborative Partnerships

Each branch is explored, pros/cons are listed, and then combined into a final hybrid strategy.



When to Use ToT



Ideal for:

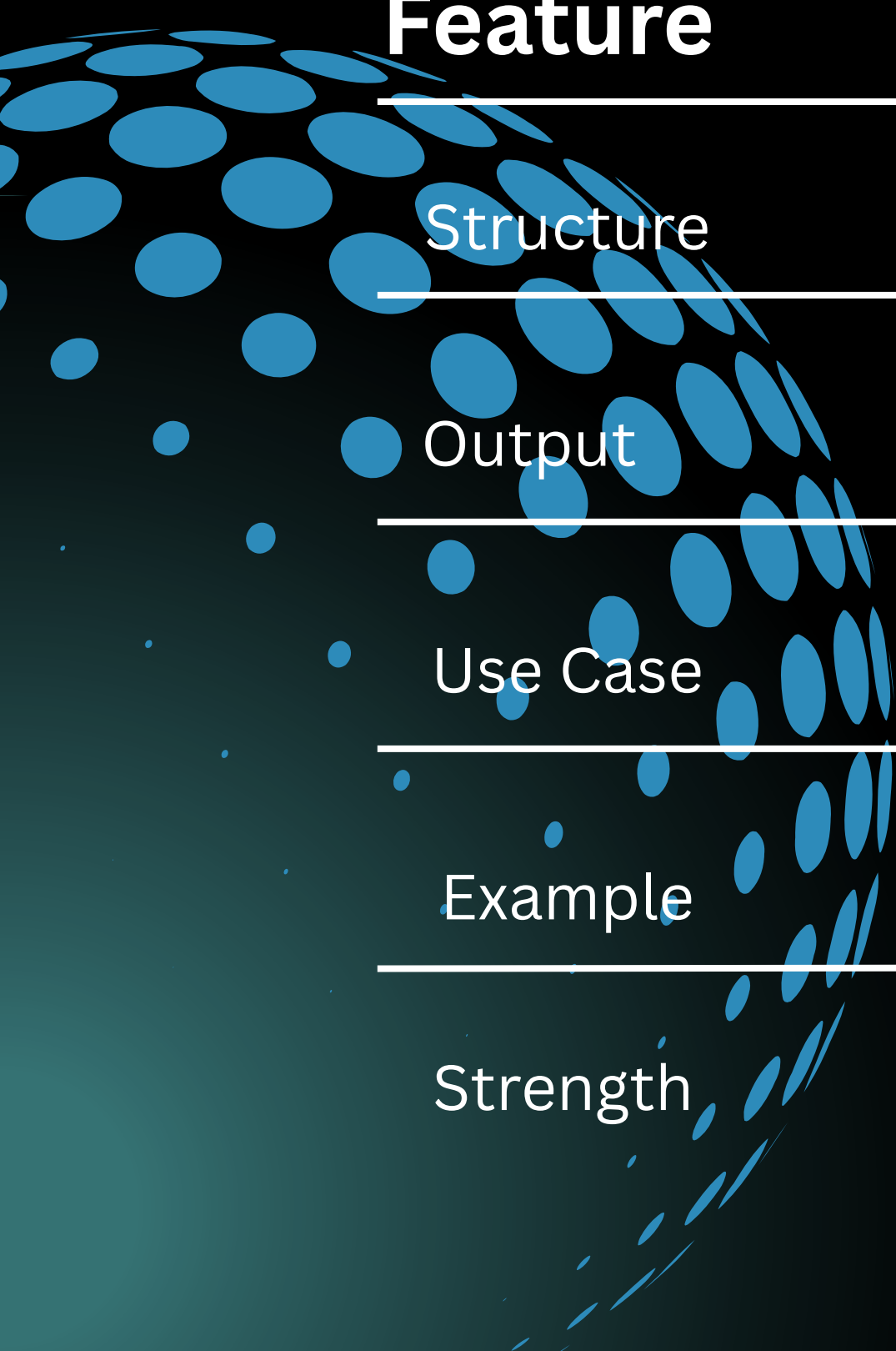
- Creative brainstorming
- Strategic planning
- Decision-making with multiple factors
- Problem solving with alternative paths.



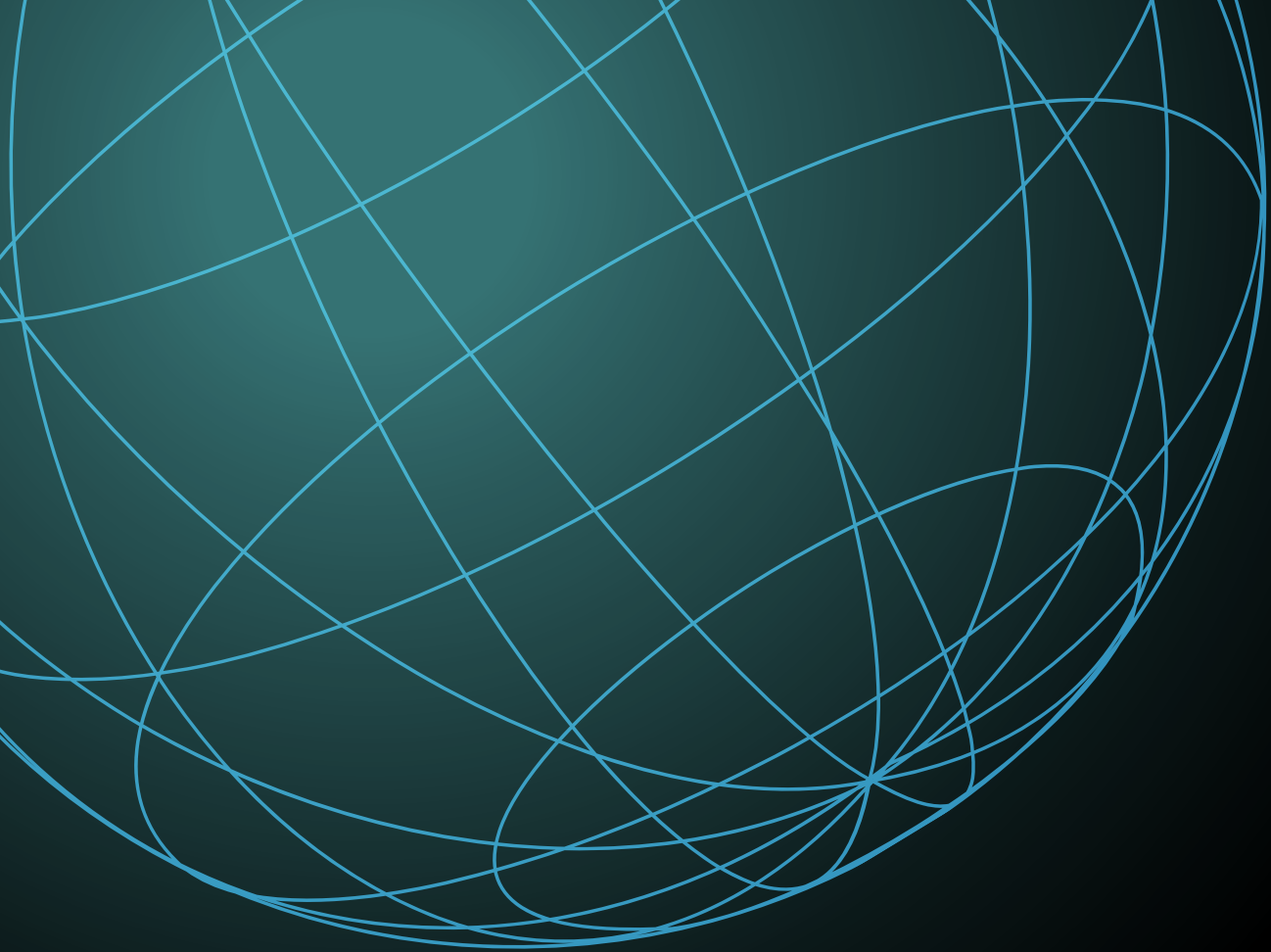
Benefits:

- Encourages exploration
- Prevents bias toward one idea
- Creates balanced, evaluated solutions

CoT vs ToT — Key Differences



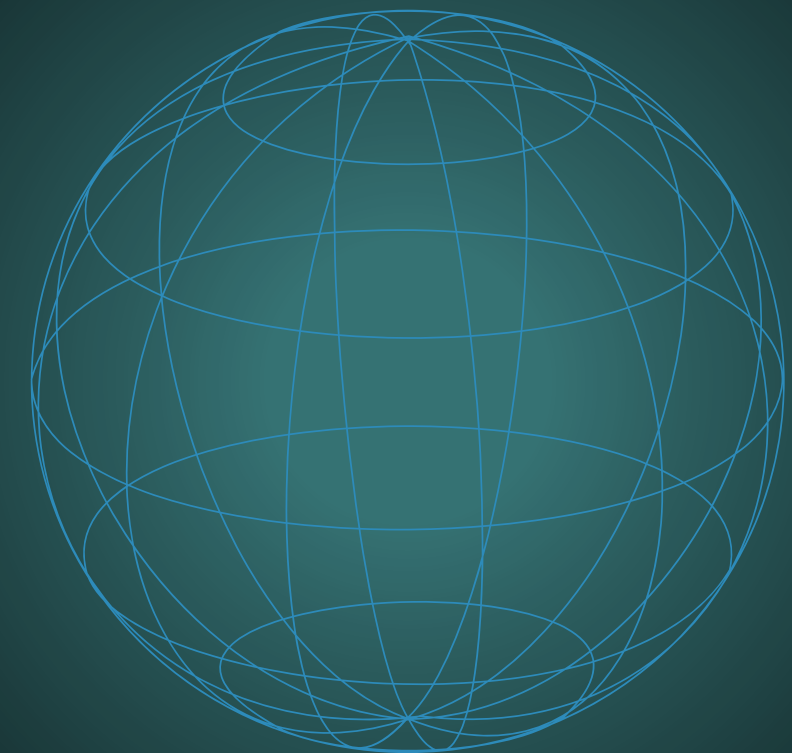
Feature	Chain of Thought (CoT)	Tree of Thoughts (ToT)
Structure	Linear reasoning	Branch-based reasoning
Output	One logical path	Multiple compared ideas
Use Case	Analytical thinking	Strategic exploration
Example	Solving math	Planning a campaign
Strength	Clear logical flow	Creative depth



Real-World Applications

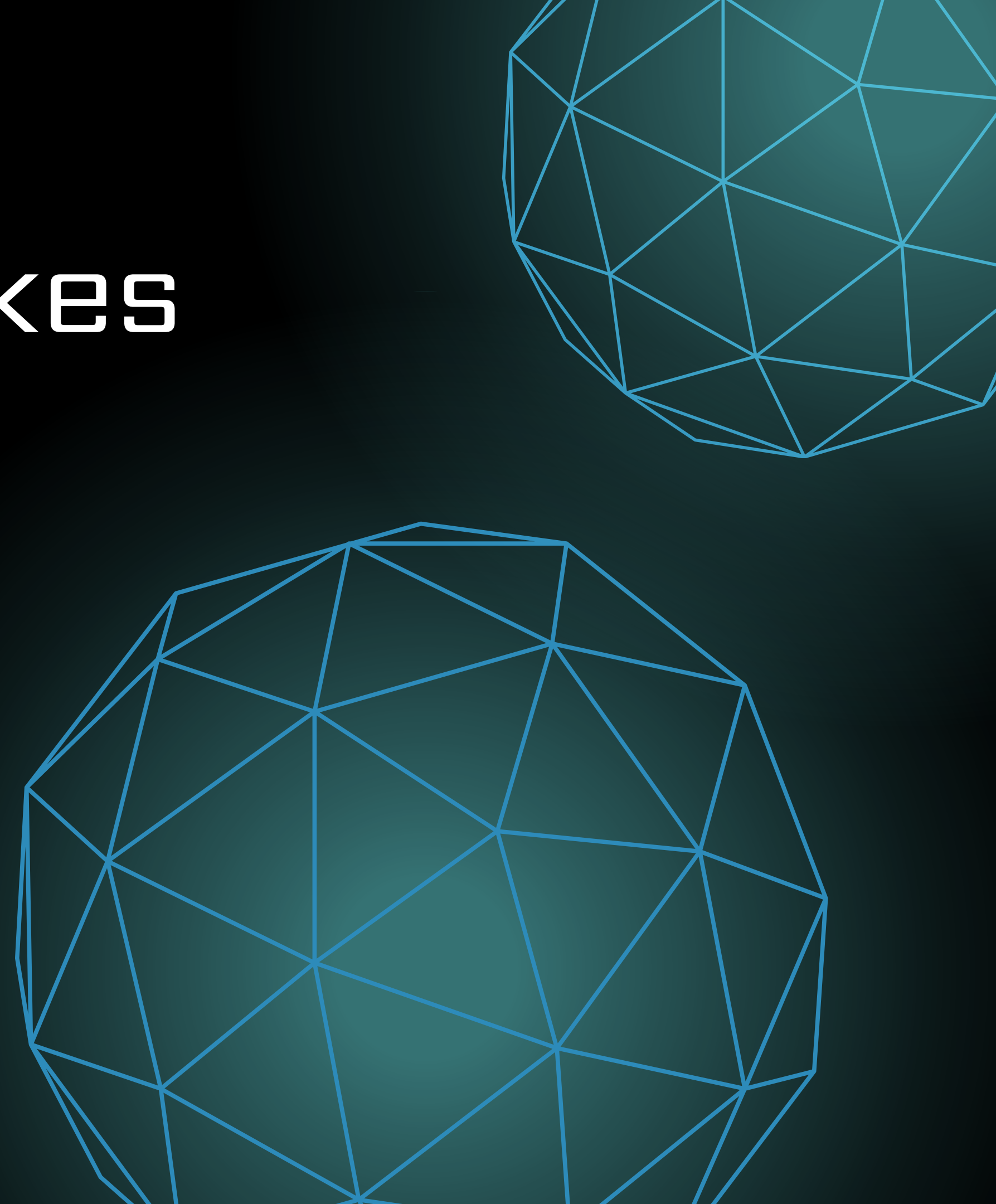
- CoT:
- Math tutoring systems
- AI coding assistants
- Logic-based Q&A models

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Common Mistakes

- CoT: Overlong reasoning, repetitive steps.
- ToT: Too many branches → confusion or high compute cost.



Combining CoT and ToT

- Start with ToT for idea generation.
 - Refine each idea using CoT for detailed reasoning.
- ✓ Example: "Brainstorm 3 ideas (ToT), then explain the best one step-by-step (CoT)."

Thank You